PROJECT DOCUMENTATION

# Global Child Mortality: Regional Inequalities and Key Drivers

A Power BI analytics project using WHO Global Health Observatory data (2000–2017) to examine under-five, infant, and neonatal mortality by region.

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# 1. Project Overview

## Purpose

This project analyses global child survival outcomes using official WHO public data. The goal is to show where under-five deaths are concentrated, how mortality differs by WHO region, and how immunization coverage and leading causes of death relate to child survival.

### Domain

Public Health &
Child Survival
(SDG 3.2)

### Focus

WHO Regions
Mortality Rates
Immunization

### Output

Interactive
Power BI Dashboard
+ Documentation

# 2. Research Problem & Objectives

### Problem Statement

Despite global progress, under-five mortality remains high and unevenly distributed. Many deaths are preventable. Decision-makers need clear regional evidence on where the burden is highest, which causes dominate, and whether routine immunization coverage aligns with survival outcomes.

## Project Objectives

1. Compare neonatal, infant, and under-five mortality across WHO regions
2. Identify regions with the highest and lowest child mortality burden
3. Examine DTP3 immunization coverage as a marker of routine health services
4. Highlight leading causes of under-five deaths (preventable vs treatable)
5. Produce a region-level Power BI dashboard for research and portfolio use

# 3. Data Source — World Health Organization

All indicators were obtained from the WHO Global Health Observatory (GHO), the official public repository for global health statistics.

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|---|---|
| **Organization** | World Health Organization (WHO) |
| **Portal** | Global Health Observatory (GHO) — who.int/data/gho |
| **Period** | 2000–2017 (analysis window used in dashboard) |
| **Geography** | Country-level data aggregated to six WHO regions |
| **Access** | Open public data (free, legal for research & portfolio use) |

# 4. Data Dictionary — Indicators Used

| Indicator | Definition (short) | Unit |
|---|---|---|
| **Neonatal mortality rate** | Deaths in first 28 days of life | Per 1,000 live births |
| **Infant mortality rate** | Deaths before age 1 | Per 1,000 live births |
| **Under-five mortality rate** | Deaths before age 5 | Per 1,000 live births |
| **Number of under-five deaths** | Estimated absolute deaths under 5 | Count |
| **DTP3 immunization coverage** | 3rd dose DTP among 1-year-olds | % |
| **Causes of death (<5 years)** | Distribution of causes among under-5 deaths | % |

# 5. Tools, Data Cleaning & Methods

| | |
|---|---|
| **Excel** | Initial inspection, column review, export preparation |
| **Power Query** | Removed unused columns, renamed fields, set data types, filtered sex/age where needed |
| **Power BI** | Data model, DAX measures, region-level dashboard design, slicers |
| **SQLite** | SQL queries for region averages and country lists |

Approach: descriptive region-level analysis. KPIs include average mortality rates, total under-five deaths, DTP3 coverage, and cause distribution. Not causal inference.

# 6. Dashboard Overview



# 7. Key Findings

## 01 Regional inequality

Under-five mortality differs substantially across WHO regions. The gap between highest and lowest regions shows uneven child survival progress.

## 02 Death burden concentration

A large share of total under-five deaths is concentrated in specific regions. Progress there has the greatest impact on the global total.

## 03 Immunization alignment

Regions with higher DTP3 coverage tend to show better child survival indicators, consistent with stronger routine health-service reach.

## 04 Preventable causes dominate

Leading causes include prematurity, birth complications, pneumonia, diarrhoea, and malaria — many are preventable or treatable.

# 8. Recommendations — Solutions to Reduce Rates

Aligned with WHO guidance on ending preventable child deaths (SDG 3.2).

### Neonatal care

Skilled birth attendance; essential newborn care; Kangaroo Mother Care; manage birth asphyxia and neonatal sepsis

### Immunization

Raise DTP3 and full immunization coverage toward 90% (Immunization Agenda 2030); close zero-dose gaps

### Pneumonia & diarrhoea

Pneumococcal & rotavirus vaccines; exclusive breastfeeding; oral rehydration + zinc; timely antibiotics

### Malaria (high-burden regions)

Insecticide-treated nets; intermittent preventive treatment; rapid diagnosis and artemisinin-based therapy

### Health system access

Primary care strengthening; antenatal care; referral systems for emergency obstetric and newborn care

### Nutrition & WASH

Address undernutrition; safe water, sanitation, and hygiene to reduce infection-related deaths

# 9. Limitations, Conclusion & Skills

## Limitations

- Not all indicators available for every country-year
- Causes-of-death series is more limited in years
- Multi-year averages can hide recent change
- Analysis is descriptive, not causal

## Conclusion

Child mortality remains regionally unequal. High-burden regions need priority investment in neonatal care, immunization, and treatment of leading infectious and perinatal causes — consistent with WHO strategies to achieve SDG 3.2.

## Skills Demonstrated

Data cleaning (Power Query) • Data modeling • DAX measures • Dashboard design
Regional health analytics • Insight communication • SQL querying
Use of official WHO open data •

# 10. References & Contact WHO

## Primary data source

World Health Organization. Global Health Observatory (GHO) data repository.
https://www.who.int/data/gho

## Key WHO references for solutions & targets

- WHO — SDG Target 3.2: end preventable deaths of newborns and children under 5
- WHO / UNICEF — Levels & Trends in Child Mortality (UN IGME reports)
- WHO — Immunization Agenda 2030 (90% coverage targets)
- WHO — Every Newborn Action Plan; integrated management of childhood illness (IMCI)

### Contact / further information (WHO)

WHO Headquarters, Geneva • https://www.who.int • Data inquiries via GHO portal
Regional offices: AFRO, AMRO/PAHO, EMRO, EURO, SEARO, WPRO
This project uses only published open data; it is not an official WHO product.